

SEPHAN OGOLA OTIENO

Full-Stack Engineer | Solutions Engineer | API Integration Specialist
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EXECUTIVE SUMMARY

Strategic and results-oriented Solutions Engineer with over 3 years of experience in architecting resilient API-driven ecosystems and high-volume data pipelines. Expert in Python and Node.js, with a deep focus on bridging the gap between fragmented SaaS platforms and automated business workflows. Proven track record in designing self-hosted automation hubs (n8n, Temporal), executing massive document migrations (300k+ records), and engineering secure, scalable infrastructures. A self-directed problem solver adept at leveraging structured thinking to build fault-tolerant systems that optimize performance and minimize operational costs in hybrid AI/Human environments.

CORE TECHNICAL SKILLS

- Languages: Python, JavaScript, TypeScript, Bash, SQL.
- Version Control System: Git
- API & Integration: REST, GraphQL, Webhooks, OAuth 2.0, Microsoft Graph API, Axios.
- Automation & Orchestration: n8n, Temporal, Redis, PostgreSQL, Elasticsearch.
- DevOps & Cloud: Docker, Coolify, NGINX, PM2, GitHub Actions, DigitalOcean, Linux (Ubuntu).
- Data Engineering: ETL Pipelines, Batch Processing, Rate Limit Management, Exponential Backoff.
- SaaS Ecosystems: CRM Integration, Airtable, Google Sheets, Notion, Messaging APIs (WhatsApp)

PROFESSIONAL EXPERIENCE (PROJECT CASE STUDIES)

1. Mass Data Pipeline & Database Synchronization (Manatal to Zoho)

Role: Solutions Architect

Project: Automated end-to-end migration and synchronization of 300,000+ documents and records using Python and Node.js scripts.

- Built a resilient two-stage Node.js data pipeline optimized for high-performance throughput and minimal latency during large-scale batch processing.
- Managed complex REST API rate limits and implemented sophisticated pagination logic to ensure uninterrupted data flow and quota compliance.
- Engineered fault-tolerant scripts using exponential backoff and automated retries to handle transient API failures and connection timeouts gracefully.
- Conducted comprehensive post-transfer verification and data cleansing to validate data integrity across 100% of the migrated JSON records.
- Deployed and supervised the pipeline using PM2 on DigitalOcean, maintaining zero data loss over a continuous 36-hour synchronization window.

2. Fault-Tolerant Social Media Automation & Messaging API Integration (Postiz)

Role: Backend & Integration Engineer

Project: Self-hosted automation ecosystem integrated with multiple Social Media REST APIs and communication webhooks.

- Designed and implemented background workflows using Temporal to manage long-running integration tasks and guarantee reliable event delivery.
- Integrated real-time Webhooks for automated response handling from third-party ad platforms, significantly reducing manual engagement through outreach automation.
- Optimized system performance to address latency issues, ensuring fast response times for high-frequency API authentication and OAuth requests.
- Leveraged Redis for high-speed caching and PostgreSQL for relational data storage to support complex, stateful data synchronization logic.

3. E-commerce API Integration & Local Payment Synchronization (Browns Pharmacy)

Role: Full-Stack Integration Specialist

Project: Catalogue synchronization across Zoho Commerce, Zoho Inventory, and local M-Pesa REST API gateways.

- Built automated data cleansing and batch import pipelines using Python and Node.js, reducing manual catalog management by over 90%.
- Integrated Safaricom M-Pesa and Pesapal REST API to synchronize real-time transaction data with internal accounting and inventory databases.
- Developed custom validation logic to ensure real-time data consistency between warehouse inventory levels and front-end product availability.
- Optimized the checkout workflow by implementing conditional logic for prescription-only items, ensuring strict regulatory compliance.
- Managed API rate limits and data exchange formats (JSON) to facilitate seamless connectivity between vendor systems and the core store.

4. Enterprise Identity Integration & Communication Workflow (Gap Recruitment)

Role: Solutions Engineer

Project: Microsoft Graph API and SharePoint integration for automated high-traffic recruitment and communication workflows.

- Built a secure PWA kiosk that connects to Azure AD for real-time authentication and cross-platform identity synchronization.
- Automated outreach and notification workflows triggered by real-time arrival events, improving stakeholder communication efficiency through Messaging APIs.
- Ensured secure data flow by implementing hardened API security protocols including Content Security Policy (CSP), HSTS, and robust failure handling.
- Developed robust error handling for Microsoft 365 API quotas and rate limits to prevent system downtime during peak recruitment periods.
- Evaluated integration between core business systems to ensure reliable data flow and consistency across the Microsoft 365 ecosystem.

5. Private PaaS Orchestration & CI/CD Deployment Pipeline (Coolify)

Role: DevOps & Infrastructure Engineer

Project: Centralized orchestration layer to automate the deployment of 10+ integration services and REST APIs on a private VPS.

- Designed and built a centralized management platform to evaluate and improve data pipeline performance and latency.
- Optimized Git-to-Deploy workflows using GitHub Actions CI/CD to ensure consistent and reliable delivery of API code updates.
- Implemented automated SSL termination, health checks, and self-healing container policies to guarantee 99.9% uptime for business-critical integrations.
- Leveraged NGINX and Traefik as reverse proxies to manage traffic flow, manage rate limits, and secure external API endpoints.
- Developed custom Bash scripts for infrastructure-as-code automation, ensuring fault-tolerant deployments for all synchronized services.

EDUCATION & CERTIFICATION

- Bachelor's Degree in Electrical and Electronic Engineering
- Certified AWS Cloud Practitioner

PROFESSIONAL INTERESTS

- Hybrid AI+Human Automation Systems & AI Pilots
- Real-time Data Pipeline & Database Synchronization Optimization
- Open-Source Integration & Connectivity Frameworks

LANGUAGES

English – Fluent | **Swahili** – Fluent | **French** – Functional